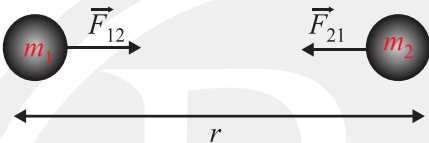
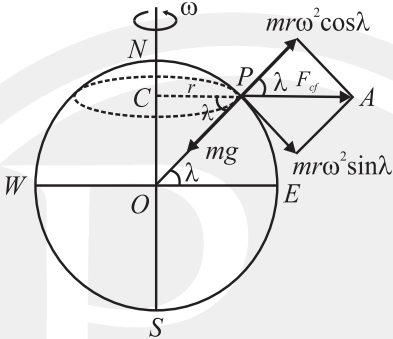
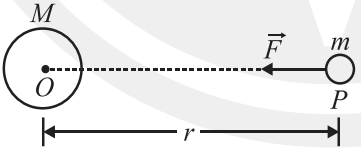
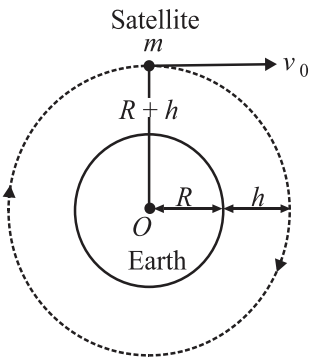


Gravitation

Sr. No.	Concept	Formulas	Other information
1.	Newton's law of gravitation	 Gravitational force acting between the bodies: $\Rightarrow F_g \propto \frac{m_1 m_2}{r^2}$ $\Rightarrow F_g = \frac{G m_1 m_2}{r^2}$ $ \vec{F}_{12} = \vec{F}_{21} = F_g$ In vector form: $\vec{F}_{12} = -G \frac{m_1 m_2}{r^2} \hat{r}_{21}$ Value of G: $6.67 \times 10^{-11} \text{ N-m}^2/\text{kg}^2$; Principle of superposition: $\vec{F}_R = \vec{F}_{01} + \vec{F}_{02} + \vec{F}_{03} + \dots \dots \dots \vec{F}_{0n} = \sum_{i=1}^n \vec{F}_{0i}$	F_g = gravitational force G = gravitational constant m_1, m_2 = mass of the two bodies respectively. r = distance between the two bodies. F_R = resultant gravitational force acting on particle due to number of point masses. $\vec{F}_{01}, \vec{F}_{02}, \dots \vec{F}_{0n}$ = Gravitational forces on a particle of mass m_0 due to particle of masses $m_1, m_2, m_3, \dots \dots \dots m_n$ respectively.
2.	Acceleration due to gravity	At the surface of the earth, $g = \frac{GM}{R^2}$ Mean density of the earth: $\rho = \frac{M}{\frac{4}{3}\pi R^3} = \frac{3g}{4\pi GR}$ Weight of a body: $W = mg$	g = acceleration due to gravity R = radius of the earth ρ = density of the earth m = mass of object on earth M = mass of earth

Sr. No.	Concept	Formulas	Other information
3.	Variation of acceleration due to gravity with height and depth	Effect of altitude: The value of g at height h from earth's surface is $g' = \frac{g}{\left(1 + \frac{h}{R}\right)^2}, \text{ if } h \ll R, \quad g' = g \left(1 - \frac{2h}{R}\right)$ Effect of depth: The value of g at depth d from earth's surface is $g' = g \left(1 - \frac{d}{R}\right)$	R = radius of the earth g = acceleration due to gravity at earth's surface.
4.	Variation of gravitational acceleration due to rotation of earth	 Acceleration at point P: $g_\lambda = g - R\omega^2 \cos^2 \lambda$ At the equator, $\lambda = 0^\circ$, $\cos \lambda = 1$, hence $g_e = g - R\omega^2$ At the poles. $\lambda = 90^\circ$, $\cos \lambda = 0$ hence $g_p = g - R\omega^2 \times 0 = g$	m = mass of the object g_λ = acceleration due to gravity at point P when rotation of earth is taken into account. λ = altitude of the place R = radius of earth ω = angular velocity of earth g = acceleration due to gravity when earth is at rest.
5.	Gravitational field intensity	 According to Newton's law of gravitation, $F = \frac{GMm}{r^2}$ $\therefore E = \frac{F}{m} = \frac{GM}{r^2}$ In vector form, $\vec{E} = -\frac{GM}{r^2} \hat{r}$	E = gravitational field intensity at point P. F = gravitational force acting between them. r = distance between two bodies.

Sr. No.	Concept	Formulas	Other information
6.	Gravitational potential and potential energy	Gravitational potential, $V = \frac{\text{Work done}}{m} = -\frac{GM}{r}$ Gravitational potential energy, $U = -\frac{GMm}{r}$ $E = -\frac{dV}{dr}$ Total energy of a body in a gravitational field, $= K.E. + P.E. = \frac{1}{2}mv^2 + \left(-\frac{GMm}{r}\right)$	U = gravitational potential energy E = gravitational field intensity m = mass of the test body v = velocity of the body r = distance of test body. M = mass of the earth.
7.	Gravitational potential of a spherical shell	If point lies inside the spherical shell ($r < R$): Gravitational potential, $V = \frac{-GM}{R}$ If point lies on the surface of the spherical shell ($r = R$): Gravitational potential, $V = \frac{-GM}{R}$. If point lies outside the spherical shell ($r > R$): Gravitational potential, $V = \frac{-GM}{r}$.	R = radius of the spherical shell. M = mass of the spherical shell. r = test point distance from centre.
8.	Gravitational potential of a solid sphere	If point lies inside the uniform solid sphere ($r < R$): The value of gravitational potential is given by, $V = -GM \left[\frac{(3R^2 - r^2)}{2R^3} \right]$ If point lies on the surface of the uniform solid sphere ($r = R$): Gravitational Potential $V = \frac{-GM}{R}$. Gravitational potential at the centre of the solid sphere is given by, $V = \left(-\frac{3}{2}\right) \times \left(\frac{GM}{R}\right)$.	R = radius of the solid sphere. M = mass of the solid sphere. r = test point distance from centre.
9.	Escape velocity	Escape velocity of a body on earth's surface: $v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR} = \sqrt{\frac{8}{3}\pi G\rho R^2}$	v_e = escape velocity on earth. ρ = density of the earth. M = mass of earth. R = radius of earth.

Sr. No.	Concept	Formulas	Other information
10.	Orbital motion of satellite	 Orbital velocity at a height h, $v_0 = \sqrt{\frac{GM}{R+h}} = \sqrt{\frac{gR^2}{R+h}} = R \sqrt{\frac{g}{R+h}}$ When a satellite revolves close to earth's surface ($h \ll R$) $v_0 = \sqrt{gR}$ $v_e = \sqrt{2}v_0$ Time period of a satellite $T = \frac{2\pi(R+h)}{v_0} = 2\pi \sqrt{\frac{(R+h)^3}{GM}} = \frac{2\pi}{R} \sqrt{\frac{(R+h)^3}{g}}$ $= \sqrt{\frac{3\pi(R+h)^3}{G\rho R^3}}$ For satellite orbiting close to the surface of earth, ($h \ll R$). $T = \sqrt{\frac{3\pi}{G\rho}}$ Height of a satellite, $h = \left[\frac{T^2 R^2 g}{4\pi^2} \right]^{1/3} - R$	M = mass of the earth. R = radius of the earth. m = mass of the satellite. v_0 = orbital velocity of the satellite. h = height of the satellite above the earth's surface. $R + h$ = orbital radius of the satellite.
11.	Angular momentum of the satellite	Angular momentum of a satellite moving with velocity v in a orbit of radius r : $L = mvr = m\sqrt{\frac{GM}{r}} \cdot r = \sqrt{GMm^2 r}$	$r = (R + h)$ position of the satellite from the centre of earth. m = mass of the satellite.

Sr. No.	Concept	Formulas	Other information
12.	Potential and kinetic energy of a satellite	Potential energy, $U = -\frac{GMm}{r}$ Kinetic energy, $K = \frac{1}{2}mv_0^2 = \frac{1}{2}m\left(\frac{GM}{r}\right) = \frac{1}{2}\frac{GMm}{r}$ Total energy, $E = K + U = \frac{1}{2}mv_0^2 - \frac{GMm}{r} = -\frac{GMm}{2r}$ Binding energy $= \frac{GMm}{2r}$	v_0 = orbital velocity of a satellite. M = mass of the earth m = mass of the satellite r = orbital radius of the satellite from centre of earth.
13.	Kepler's law of planetary motion	Law of areas, $\frac{\Delta A}{\Delta t} = \text{constant}$ Law of periods, $T^2 \propto r^3$ or $T^2 = kr^3$. For a satellite of earth, $k = \frac{4\pi^2}{GM_E} = 10^{-13} \text{ s}^2 \text{ m}^{-3}$ $\frac{T_2^2}{T_1^2} = \frac{r_2^3}{r_1^3}$	T = time period of revolution of planet r = length of the semi major axis of elliptical orbit. ΔA = area swept out in time interval Δt